Made By - Karan Agrawal (GATE CS 2024 AIR 102)

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First Order Logic: Also known as

→ Predicate Calculus
→ Predicate Logic
→ Quantificational Logic
→ First Order Logic (FOL)

As we already had Prop. Logic, then why did we need another logic. Actually prop. logic was world of True and false. Not suitable for all statements like this:

Ex: All men are Mortal.
Socrates is a men.
∴ Socrates is mortal.

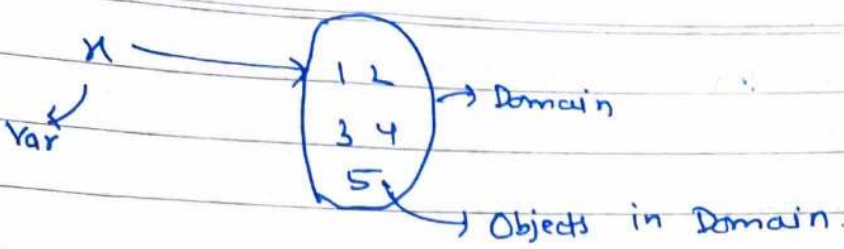
This statement cannot be proved valid via prop. logic. because in PL, these 3 statements are d/f atomic propositions.

So we can say PL has very limited expressive power and that is why we need new ML.

Main issue with PL is that it treats each var. as True or false only.

→ First Order logic is a world of objects, their properties, their transformation (function). FOL has:

a) In this each variable refers to some object of its domain (or universe). In PL each var. referred to either True or False.

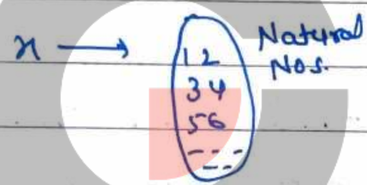


→ Each var. has a domain and d/f variables can have different domains. Each var. can refer to object from its own domain only.

→ Objects are constants which are elements in the domain.

b). FOL is concerned about properties of these Objects.

Ex:



Even(x) : x is Even

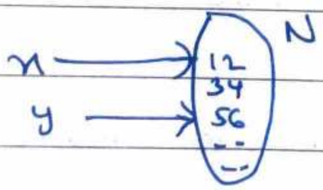
Even(2) : True

Even(3) : False

Even(-1) : Invalid bcz -1 is not in domain of x.

c). In FOL, we have relations among objects (also known as predicates).

Ex:



P(x,y) : x<y

P(1,2) : 1<2 = True

P(3,4) : 3<4 = True

P(5,4) : 5<4 = False

P(-1,2) = Invalid

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Ex:2 Sum (x, y, z) : x+y=z

sum (2,3,5) : 2+3=5 = True

sum (1,3,40) : 1+3=40 = False

sum (0,1,1) : 0+1=1 Invalid as 0 is not in domain.

sum (1,1,2) : 1+1=2 = True

So, predicates can be over 0,1,2 --- n variables.

→ Predicate over 0 var is just proposition.

Ex: P: Ram is cool guy.

→ Predicate over 1 var. are also known as properties.

So, properties ≡ unary predicates

Ex: odd(x): x is odd.

→ Once we replace every variable in a predicate with some value from their domain, then it becomes proposition bcz value is True or False.

Ex: sum (x,y,z) : x+y=z

sum (1,2,3) : $\underbrace{1+2=3}$ → This a prop. with truth value True

→ Predicate with n variables is called nary predicate.

d). Another significant new concept in FOL is quantification i.e. the ability to assert that a particular property holds for all elements or that it holds for some elements.

→ In mathematical logic some means atleast one and all means all.

$\in$ → belongs to | Element of

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- **for all ($\forall$):**

Ex: $x \rightarrow \underbrace{a, b, c, d}_{D(x)}$

$\forall_{x \in D(x)} \text{young}(x) = \forall_x \text{young}(x)$

It is understood that all x from its domain only. So no need to write it explicitly.

Above line basically means all x are Young.
For above statement to be true

young(a) $\wedge$ young(b) $\wedge$ young(c) $\wedge$ young(d) = True

So $\forall_x P(x)$ is read as for all x (in its domain) [implicit], $P(x)$ is True.

We can give any one counter example to prove it false

Ex: $P(x)$: x is even
domain(x): Natural Nos.

$\forall_x P(x)$ = false
Counter example 3 i.e. $P(3)$ = false.

- **There Exist ($\exists$): Some / Atleast One**

It is read as $\exists_x P(x)$, atleast one x is there or there exist an x (in its domain), $P(x)$ is True.

For it to be true:

$P(a) \vee P(b) \vee P(c) \ldots \vee P(n)$ = True

↳ Atleast one must be True.

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We can give any one withness to prove it true.

Ex: $P(x)$= x is Even.

domain(x) = Natural Nos

$\exists x\, P(x)$ = True

withness = 2 i.e. $P(2)$ is True

Q. In FOL, we also have functions of objects.

Ex: $f(x) : x^2$

$f(2) = 4$

(will talk later)

FOL vs PL:

→ PL assumes that there are fact which either hold or do not hold. Whereas FOL assumes that there are:

- Objects: people, colors, number, facts, games, etc.
- relations: is red, round, prime, bigger than, etc.
- functions: father of, one plus, plus, etc.

→ FOL is superset of PL.

(That was some intro to FOL, now we will go in detail of this topic).

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First Order Logic Contains:

- Objects : Domain
- Properties of Objects : Predicates
- Talk about multiple Objects : Quantifiers

→ As PL was world of facts (T/F), FOL is just like Real world (People & their stories) i.e Objects & their properties.

↓ properties: Predicate

↓ Objects: Domain

→ Domain / Universe: The set of objects we discuss is called Domain.

Ex: Set of Real nos.

Set of Peoples.

- Variable can take value from its domain only.
- Domain can be any non-empty set at all. Any set.
- Objects in the domain are also called the Constants.

→ Predicate: It tells us about the properties of Objects and relationship among objects.

Ex:) Domain: set of people

male(x): x is a male.

↳ Property

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$f(x,y)$: x is father of y.

→ Relationship among objects.

→ Predicate is a sentence containing variables (such that every variable refers to its own domain obviously), such that it becomes a proposition once we replace each variable with specific value from domain.

Ex: $x>3$, $x=y+3$, $x+y=2$

$x>3$ → Unary predicate → Property

$x=y+3$ → Binary predicate, $x+y=2$ → Ternary predicate → Relationship.

→ Predicates are also known as propositional functions as predicates become proposition when we replace every var. by constant from domain of that variable.

NOTE Domain in FOL is always non-empty ; unless explicitly stated. By default, domain of every variable in a FOL expression is same and is called domain of expression.

⇒ There are two ways to make propositions from predicates

→ Way 1: Substitute every variable by some value from the domain.

Ex: Domain : $\mathbb{Z}$

$P(x,y) : x=y+3$

$P(5,0) : 5=0+3$ → Proposition with TV false.

Quantification: Satisfaction by multiple objects

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→ **Way 2: Quantification:**

Ex: Domain: $\mathbb{Z}$
$E(x)$: x is Even
for all x, $E(x)$ is True
↳ Proposition with Truth val False.

we will go in more detail of quantification further.

⇒ So, a predicate becomes a prop. when we assign it fixed values or when we quantify it.

☆ There are various quantification words in English:
• Few • All • Many • A little • Most • Some • No

But, in mathematics we have two quantification words only:

☆ 1) **Universal Quantification:** It means saying that a property P is satisfied by ALL elements in the domain (or universe).
Repr. by "$\forall$". Read as "for all".

$\forall x \in D \; P(x) \equiv \boxed{\forall x \; P(x)}$ because $x \in D$ is understood.

Above line can be read as:
→ All elements in domain of x satisfy property P.
→ for all x, $P(x)$.

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So $\forall x\, P(x)$ is True iff $P(x)$ is True for every choice of x.

We can give one counter example (element for which $P(x)$ is false) to say that $\forall x\, P(x)$ is false.

Ex: Domain = Real Nos.

a) $P(x)$: $x \geq 0$

$\Rightarrow$ $\forall x\, P(x)$ is false, counter example $x = -1$

b) $q(x)$: $x^2 \geq 0$

$\Rightarrow$ $\forall x\, q(x)$ is True. No counter example.

c) $r(x)$: $x^2 - 3 > 0$

$\Rightarrow$ $\forall x\, r(x)$ is false, counter example is $x = 1$

→ An element for which $P(x)$ is false is called counterexample for $\forall x\, P(x)$.

→ Universal quantifier when domain is finite set acts like a conjunction (and)

Ex: if domain = $\{a, b, c, d\}$

then $\forall x\, P(x) = P(a) \wedge P(b) \wedge P(c) \wedge P(d)$

i.e. all must be True.

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★ b) **Existential Quantification**: It means saying that a property is satisfied by atleast one element in the domain.

Some ≡ At least one.

Repr. by $\exists$, read as "There Exists".

$\exists_{x \in D} P(x) \equiv \boxed{\exists x\, P(x)}$

Above line can be read as:

→ At least one element satisfy $P(x)$

→ There exist an element x in the domain, $P(x)$ is True.

→ There exist x, $P(x)$.

→ Some element in domain satisfy property P.

So $\exists x\, P(x)$ is True if ~~at least~~ $P(x)$ is true for atleast one choice of x.

We can given any one witness (an element for which $P(x)$ is True) to prove $\exists x\, P(x)$ True.

Ex: Domain = Real Nos.

a) $P(x): x \geq 0$

$\Rightarrow$ $\exists x\, P(x)$ is True. , witness = 1

b) $q(x): x^2 \geq 0$

$\Rightarrow$ $\exists x\, q(x)$ is True, witness = 2

c) $r(x): x^2 = -1$

$\Rightarrow$ $\exists x\, r(x)$ is False, No witness.

→ Existential quantifier when domain is finite set acts like a disjunction (or).

Ex: domain = $\{a, b, c, d\}$

$\exists x\, P(x) = P(a) \lor P(b) \lor P(c) \lor P(d)$

i.e. atleast one must be True.

Que: Determine truth value of following statements if the domain consists of all integers

a) $\forall n\, (n+1 > n)$

→ True, (Obvious bcz all $n+1$ are greater than n).

b) $\exists n\, (2n = 3n)$

→ True, (witness = 0)

c) $\exists n\, (n = -n)$

→ True, (witness = 0)

d) $\forall n\, (3n \leq 4n)$

→ False (counter ex = -1)

Note: $\forall n\, \boxed{\text{exp.}}$, means, all n should be true for this expression.

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Que: Suppose $P(x)$ is predicate $x+2 = 2x$ & the universe for x is the set $\{1, 2, 3\}$. Then:

a) $\forall x\, P(x)$
→ False, (counterex: 1)

b) $\exists x\, P(x)$
→ True, (~~counter~~ witness = 2)

c) $\forall x\, \neg P(x)$
→ It means for all x $\neg P(x)$ should be true.
i.e. $P(x)$ should be false for all x, but $P(x)$ is true for $x=2$, so counter ex: 2
False

d) $\exists x\, \neg P(x)$
→ It means $\neg P(x)$ is True for some x, i.e. $P(x)$ is false for some x, i.e. True, (witness = 1)

e) $\neg \exists x\, P(x)$
→ It means, there do not exist any $P(x)$ which is True, which is False (~~witness~~ (witness, $x = 2$)

f) $\neg \forall x\, P(x)$
→ It means, it is not the case that all $P(x)$ are true, which is True bcz. $P(x)$ is false for $x=1$.

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Que: Universe here comprises of all real numbers. Open stmts are:

$p(x): x \geq 0$ $r(x): x^2 - 3x - 4 = 0$

$q(x): x^2 \geq 0$ $s(x): x^2 - 3 > 0$

a). $\exists x\, (p(x) \wedge r(x))$

$\Rightarrow$ True. (witness : 4)

b) $\forall x\, (p(x) \rightarrow q(x))$

$\Rightarrow$ True, as if $x \geq 0$ then obviously $x^2 \geq 0$.

★ Some tricky points:

NOTE: In FOL, domain is always non-empty, unless it is explicitly given as empty.

A). Behaviour of quantifiers if domain of x is given empty:

→ $\forall x\, P(x)$ = True, bcz no counterexample.
P(x) is True for all x. (false only if Counter ex.)

→ $\exists x\, P(x)$ = False, bcz no witness
P(x) is not True, even for one x.
(True only if witness)

B). Behaviour of quantifier when no free var in proposition:

→ $\forall x\, P \equiv P$

→ $\exists x\, P \equiv P$

Ex: $P \Rightarrow 2+2=4 \rightarrow T$, $q \Rightarrow 2+2=5 \rightarrow F$

$\exists x\, p \equiv p$ = True $\qquad \forall x\, p \equiv p$ = True

$\exists x\, q \equiv q$ = false $\qquad \forall x\, q \equiv q$ = false.

→ we already know, predicate w/o var. is a proposition.

Imp.

English - FOL Translation: for this concept let

→ domain: set of all animals.

→ cute(x) : x is cute. $\equiv c(x)$

a) Every animal is cute.

$\forall x.\ cute(x)$

b) Some animal is cute.

$\exists x\ c(x)$

c) Every Rabbit is cute.

$\forall x\ (rabbit(x) \rightarrow cute(x))$

i.e. for all x if x is rabbit, then it must be cute, else it does not matter.

Note: $\forall x\ (rabbit(x) \wedge cute(x))$ means x is rabbit and x is cute, so it will be false even if all rabbits are cute but lion is uncute.

Also $\forall x\ ((cute(x)))$ is not correct bcz, it will be also false if any animal is uncute.

d) Some rabbit is cute

→ $\exists x\ (cute(x) \wedge rabbit(x))$

i.e. there is some x which is rabbit & cute.

Note. $\exists x\ (rabbit(x) \rightarrow cute(x))$ is not correct bcz it ~~means~~ will be true even if all rabbits are uncute, but some animal is not rabbit.

So in general.

All P's are Q's translates to $\forall x\ P(x) \rightarrow Q(x)$

Some P is a Q. translate to $\exists x\ P(x) \wedge Q(x)$

Qu: Translate to English:

a) $\forall x\ (R(x) \rightarrow cute(x))$

→ Every rabbit is cute.

b) $\forall x\ (R(x) \wedge C(x))$

→ Every animal is Rabbit & cute.

c) $\exists x (R(x) \rightarrow C(x))$

→ $\equiv \exists x\ (\neg R(x) \vee C(x))$

Some animal is either not rabbit or is cute.

d) $\exists x\ (R(x) \wedge C(x))$

→ Some rabbit hops.

e) Some rabbit is not cute

→ $\exists x\ (Rabbit(x) \wedge \neg(cute(x)))$

i.e. there is a rabbit which is not cute.

f) All rabbits are non-cute.

→ $\forall x\,(\text{rabbit}(x) \rightarrow \neg\text{cute}(x))$

i.e. for all animals, if it is rabbit, then it is not cute.

g) It is not the case, that some rabbit is cute.

→ $\neg\,\exists x\,(\text{rabbit}(x) \wedge \text{cute}(x))$. (eq. to f)

h) It is not the case, that all rabbits are cute.

→ $\neg\,\forall x\,(\text{rabbit}(x) \rightarrow \text{cute}(x))$. (eq. to e)

⇒ f & g are equivalent. and e & h are equivalent

★ So, in general

No P's are Q's translates as $\forall x\,(P(x) \rightarrow \neg Q(x))$

Some P's aren't Q's translates to $\exists x\,(P(x) \wedge \neg Q(x))$

Qu: Domain is set of all people then translate to FOL.

a) only males are in army.

→ $\forall x\,(\text{army}(x) \rightarrow \text{male}(x))$ or

$\forall x\,(\neg\text{male}(x) \rightarrow \neg\text{army}(x))$ (contrapos.)

★ Only P's are Q's translate to $\forall x\,(\neg P(x) \rightarrow \neg Q(x))$ or

$\forall x\,(Q(x) \rightarrow P(x))$

b) All males are in army

→ $\forall x\,(\text{male}(x) \rightarrow \text{army}(x))$ /

$\forall x\,(\neg\text{army}(x) \rightarrow \neg\text{male}(x))$

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$\forall$ is usually paired with $\rightarrow$

& $\exists$ " " " " $\wedge$.

i) All & only males are in army.

$\rightarrow$ $\forall x\ (male(x) \leftrightarrow army(x))$.

ii) Only rabbits are cute.

$\rightarrow$ $\forall x\ (cute(x) \rightarrow rabbit(x))$ or

$\forall x\ (\neg rabbit(x) \rightarrow \neg cute(x))$

j). All & only rabbits are cute

$\rightarrow$ $\forall x\ [(cute(x) \rightarrow rabbit(x)) \wedge (rabbit(x) \rightarrow cute(x))]$ or

$\forall x\ (cute(x) \leftrightarrow rabbit(x))$.

Qu: Translate "Every students in this class has taken java" to FOL.

$\rightarrow$ Case I: If domain is students of this class

$\forall x\ (java(x))$

Case II: If domain is all people.

$\forall x\ (student(x) \rightarrow java(x))$.

b) Every student has visited mexico or canada

$\rightarrow$ $\forall x\ (student(x) \rightarrow (canada(x) \vee mexico(x)))$

c). For any natural no. n, n is even iff n^2 is even

$\rightarrow$ $\forall n\ (natural(n) \rightarrow (even(n) \leftrightarrow even(n^2)))$.

If $\forall_n P(n)$ is True then obviously $\exists_n P(n)$ will be True

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★ $\forall_{p(x)} (q(x)) \equiv \forall_x (p(x) \rightarrow q(x))$

$\exists_{p(x)} (q(x)) \equiv \exists_x (p(x) \wedge q(x))$

Ex: a) $\forall_{x<0} (x^2 > 0)$

i.e. for all $x < 0$, $x^2 > 0$

$\equiv$ for all x, if $x < 0$, $x^2 > 0$

$\equiv \forall x (x < 0 \rightarrow x^2 > 0)$ (True)

b) $\exists_{y>0} (y^2 = 2)$

i.e., for some $y > 0$, $y^2 = 2$.

$\equiv$ for some y, $y > 0$ and $y^2 = 2$

$\equiv \exists y (y > 0 \wedge y^2 = 2)$ (True)

★ **Bounded and Free Variables:**

a). **Bounded Variable / Quantified Variable / Dummy Variable:**

Ex: Every natural no. is even. (Domain Natural No.)

$\forall_n \text{Even}(n)$

Here n is bounded var. or dummy var. because in english or in real, it doesn't have any variable.

Ex:2 Domain: {a, b, c}

Predicate, $N(x)$: x is Nice

$\forall_x N(x) \equiv N(a) \wedge N(b) \wedge N(c)$

and a, b, c are constant so, no free var. x is dummy variable.

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Ex:3 $\exists_x N(x) = N(a) \vee N(b) \vee N(c)$.
x here is also dummy variable.

Also $\forall_x P(x) \equiv \forall_y P(y) \equiv \forall_k P(k)$
bcz, x, y, k all are dummy. So all are same.

There is no individual importance of Bounded Variable, they are just placeholders.

b) Free Variable / Non-quantified Variable / Real Variable :

Ex: $E(x) = x$ is even
here x is real/free variable.

Ex 2: $N(x) =$ x is nice

$N(a) =$ a is nice → Proposition

free variable is free to take any value from domain.

Ex: Domain: Set of all Integers
$P(m,n)$: $m+n > 4$

a) $P(1,2)$: $1+2>4$
Proposition (False)

b) $P(9,-3)$: $9-3>4$
Proposition (True)

c) $P(0,n)$: $0+n>4$
n is free var.

d) $\forall_n P(0,n)$: $\forall_n (n>4)$
n is bound var.
Proposition (False)

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Note: If an expression contains No free variable, then it is a proposition. (and vice versa)

So, to make a prop., remove free variable either by using constant or by using quantification.

S: $\forall x\,(x>y)$ is not a prop. bcz free var. y is present.

let $S(y) = \forall x\,(x>y)$

then $\forall y\,S(y)$ is a prop. (false)

or $\forall x\,(x>10)$ is a prop. (false)

also $\exists y\,S(y)$ is a prop. (false)

Note: If A doesn't have any free variable x, then
$\forall x\, A \equiv A$ & $\exists x\, A \equiv A$.

Ex: $\forall x\,(2+2=4) \equiv 2+2=4 \equiv$ True.
↳ No free var x.

☆ Scope of a quantifier / quantified variable: The part of a logical expression to which a quantifier is called its scope.

Ex: $\forall x\ \underbrace{[\qquad\qquad]}_{\text{Scope of } \forall x}$

$\forall x\,[Hx \wedge Px]$
↳ Both are quantified x.

$\forall x\,[Hx]\ Px$
↳ Free x
↳ Bounded x.

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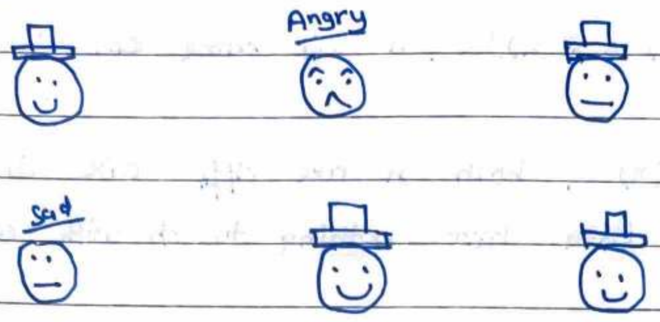
Scope plays an important role in the meaning / interpretation of an Expression:

Ex: Domain.



→ $\forall x\,(smile(x) \rightarrow wearhat(x))$ ∴ True

→ $\underset{F}{\forall x\,(smile(x))} \underset{\rightarrow}{\rightarrow} \underset{F}{\forall x\,(wearhat(x))}$ ∴ True

Both x here are diff. eq. to $\forall x\,(smile(x)) \rightarrow \forall y\,(wearhat(y))$

→ $\exists x\,(smile(x) \wedge wearhat(x))$ ∴ True

→ $\underset{T}{\exists x\,(smile(x))} \wedge \underset{T}{\exists y\,(wearhat(y))}$ ∴ True.

→ $\exists x\,(angry(x) \wedge wearhat(x))$ ∴ False.

→ $\exists x\,(angry(x)) \wedge \exists y\,(wearhat(y))$ ∴ True

So this shows us the importance of scope.

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→ In the expression $H(x) \wedge P(x)$, x is same and can also be written as $G(x) = H(x) \wedge P(x)$, (x is free) (Like in x^2+x+2 , x is same)

→ Also in $\forall x\,[H(x) \wedge P(x)]$, x is same and is bounded.

→ In $\forall x\,[H(x)] \wedge P(x)$, both x are d/f, one is dummy & other is free. Both have nothing to do with each other.

→ Also. $\forall x\,[H(x)] \wedge \forall x\,[P(x)] \equiv \forall y\,[H(y)] \wedge \forall z\,[P(z)]$
i.e. both are independent & dummy.

→ Proposition iff no free variable ($\leftrightarrow$).

→ $\forall x\,[\exists y\,(R(x,y) \wedge T(z,x)) \rightarrow \forall z\,(N(z,x,y))]$

x: B, y: B, z: F, x: B, z: B, x: B, y: F

→ $\exists x\,(x E y \wedge \exists y\,(\neg y E x))$

x: B, y: F, y: B, x: B

<u>Qu</u>: For following expressions, determine free variable. Also truth value if free variable is 0.

a) $\exists n\,(x > 2^n)$

→ Free var $= x$

$\exists n\,(0 > 2^n)$

No, n exist so that $0 > 2^n$. So <u>False</u>.

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b) $\exists n \, (n > 2^n)$

→ free var = n

$\exists n \, (n > 2^0)$

True (witness = 2 is $2 > 2^0$)

☆ Nested Quantifiers: Two quantifiers are nested if one is within the scope of other.

Ex: $R = \underbrace{\forall x \underbrace{\exists y \, \overbrace{(x + y = 0)}^{P(x,y)}}_{Q(x)}}_{R}$

$Q(x) = \exists y (x+y=0)$, here x is free var & y is bounded.

$P(x,y) = (x+y=0)$, here x & y are free var.

$R = \forall x \exists y (x+y=0)$, both x & y are bound var and R is Proposition.

→ Many interesting statements in FOL require a combination of quantifiers.

Ex: for every natural no., there exists a greater natural number ($\forall x \exists y$)

→ Quantifiers are always read from left to right.

Ex: $\forall x \exists y \, P(x,y)$

= for all x, there is some y, so that $P(x,y)$ is True

→ The four standard templates of nested quantifiers are:

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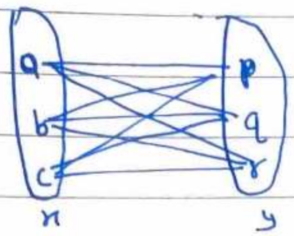
domain of x: $\{a, b, c\}$

domain of y: $\{p, q, r\}$

a). $\forall x \forall y \; p(x, y)$

- True if $p(x,y)$ is true for every pair x, y.
- False if there is a pair x, y, for which $p(x,y)$ is False



i.e. $P(a,p) \wedge P(b,p) \wedge P(c,p) \wedge$
$P(a,q) \wedge P(b,q) \wedge P(c,q) \wedge$
$P(a,r) \wedge P(b,r) \wedge P(c,r)$

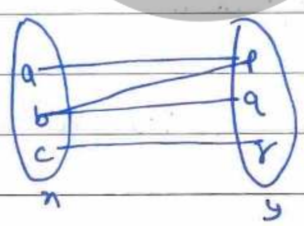
Ex: $\forall x \forall y \; (x+y == y+x)$, $x, y \in N$

True

for each pair of natural no., $a+b = b+a$.

b). $\forall x \exists y \; P(x, y)$

- True if, for every x, there is a y, so that $p(x,y)$ is True
- false if, there is an x, so that $p(x,y)$ is false for every y



i.e. $(p(a,p) \vee p(a,q) \vee p(a,r)) \wedge$
$(p(b,p) \vee p(b,q) \vee p(b,r)) \wedge$
$(p(c,p) \vee p(c,q) \vee p(c,r))$

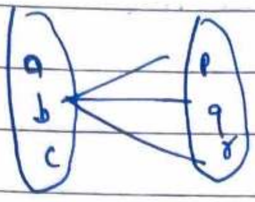
Ex: $\forall x \exists y \; (x < y)$, $x, y \in Z$ (or N)

True

for each integer x, there is some integer y, such that $x < y$

$\equiv$ for every int. there is a greater integer.

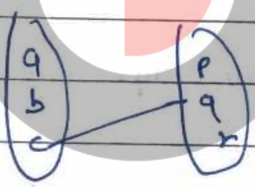
c) $\exists x \, \forall y \, P(x,y)$



- True if there is an x, for which $P(x,y)$ is True for each y.
- False if, for each x, there is a y for which $P(x,y)$ is false.

i.e. $(P(a,p) \wedge P(a,q) \wedge P(a,r)) \vee$
$(P(b,p) \wedge P(b,q) \wedge P(b,r)) \vee$
$(P(c,p) \wedge P(c,q) \wedge P(c,r))$

d) $\exists x \, \exists y \, P(x,y)$



- True, if, there is a pair x, y, for which $P(x,y)$ is True.
- False, if $P(x,y)$ is false for each pair x, y.

i.e. $P(a,p) \vee P(a,q) \vee P(a,r) \vee$
$P(b,p) \vee P(b,q) \vee P(b,r) \vee$
$P(c,p) \vee P(c,q) \vee P(c,r)$

Que: Let $P(x,y)$ denote $x+y=17$, & let U be the set of Integers then find the truth value of:

a) $\forall x \, \forall y \, P(x,y)$

→ False (Counter Ex $P(10,1)$

b) $\forall x \, \exists y \, P(x,y)$

→ True. (For each x there is some y i.e. $17-x$, so that $x+y=17$)

c) $\exists x \forall y \; p(x,y)$
→ False. (No x exist such that each y is True)

d). $\exists x \exists y \; p(x,y)$
→ True (witness $p(0,17)$)

Ex: Let domain of x : set of students.
domain of y : set of all courses
$t(x,y)$: Student x has taken course y.

then.

a) $\forall x \forall y \; t(x,y)$, means, each student has taken each course

b) $\forall x \exists y \; t(x,y)$, means, each student has taken some course

c) $\exists x \forall y \; t(x,y)$, means, some student has taken each course

d) $\exists x \exists y \; t(x,y)$, means, some student has taken some Course.

e) $\forall y \forall x \; t(x,y)$, means each course is taken by each student
(eq. to a))

f) $\forall y \exists x \; t(x,y)$, means, each course is taken by some student

g) $\exists y \forall x \; t(x,y)$, means, some course is taken by each student

h) $\exists y \exists x \; t(x,y)$, means, Some course is taken by some student
(eq. to d))

→ So, $\forall x \forall y\, P(x,y) \equiv \forall y \forall x\, P(x,y)$

$\exists x \exists y\, P(x,y) \equiv \exists y \exists x\, P(x,y)$

$\forall x \exists y\, P(x,y) \not\equiv \forall y \exists x\, P(x,y) \not\equiv \exists y \forall x\, P(x,y) \not\equiv \exists x \forall y\, P(x,y)$

Qu:

Domain: Integers

$P(x,y)$: $x < y$.

a) $\forall x \exists y\, P(x,y)$ True

b) $\exists x \forall y\, P(x,y)$ False

→ Order matters when mixing existential & universal quantifiers (Ex. $\exists x \forall y \alpha \not\equiv \forall y \exists x \alpha$), but it (order) can be changed when using any one type of quantifier (Ex: $\exists x \exists y \equiv \exists y \exists x$).

Qu:

U = real no.

$P(x,y)$: $xy = 0$

a) $\forall x \forall y\, P(x,y)$ False (Counter ex. (1,2))

b) $\forall x \exists y\, P(x,y)$ True (witness. (x, 0)) any x

c) $\exists x \forall y\, P(x,y)$ True. (witness (0, y)) any y

d) $\exists x \exists y\, P(x,y)$ True. (witness (1,0))

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Qus: U = real numbers

$p(x,y)$: $x/y = 1$

a) $\forall x \forall y \; p(x,y)$ False (Counter Ex. $P(2,3)$)

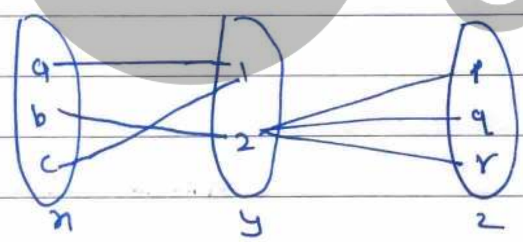
b) $\forall x \exists y \; p(x,y)$ False (No y for 0)

c) $\exists x \forall y \; p(x,y)$ False (No such x)

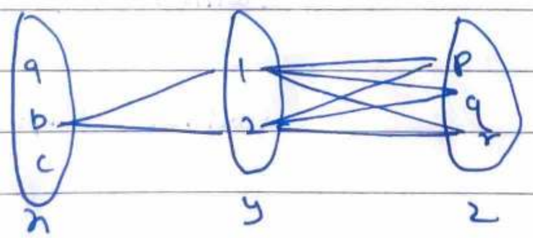
d) $\exists x \exists y \; p(x,y)$ True (witness: $p(2,2)$)

→: Nested quantifiers for more than 2 quantifiers are also there!

Ex: $\forall x \exists y \forall z \; p(x,y,z)$



Ex: $\exists x \forall y \forall z \; p(x,y,z)$



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Que: $p(x,y,z) = x+y=z$
domain : Real no.

a) $\forall x \forall y \forall z \; p(x,y,z)$
→ False (Counter ex: $p(1,3,100)$)

b) $\forall x \; \exists y \; \forall z$
→ False

c) $\forall x \; \forall y \; \exists z$
→ True.

d) $\exists x \; \forall y \; \exists z$
→ True.

Que: Determine truth value, domain is real numbers

a) $\exists x \; \forall y \; (xy=0)$
→ True $(x=0)$

b) $\forall x \; \forall y \; \exists z \; (z=(x-y)/3)$
→ True.

c) $\forall x \; \forall y \; (xy=yx)$
→ True

d) $\exists x \; \exists y \; \exists z \; (x^2+y^2=z^2)$
→ True $((3,4,5))$

→ We can think of quantification as loops also (google it).

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★ Numerical quantification (english to FOL)

→ Every cube is large

$\forall x\ (cube(x) \rightarrow large(x))$

→ No cube is large.

$\forall x\ (cube(x) \rightarrow \neg large(x))$

$\neg \exists x\ ((cube(x) \wedge large(x))$

→ Nothing is large.

$\neg \exists x\ (large(x))$

$\forall x\ (\neg large(x))$

→ Some cube is large

$\exists x\ (cube(x) \wedge large(x))$

→ All large cubes are nice.

$\forall x\ [(large(x) \wedge cube(x)) \rightarrow nice(x)]$

→ There is at least one cube

$\exists x\ cube(x)$

→ There are at least two cubes

$\exists x \exists y\ [cube(x) \wedge cube(y) \wedge x \neq y]$

↳ Req. unless True even for one cube

→ There are atleast two elements

$\exists x \exists y\ (x \neq y)$

→ There are at least three elements

$\exists x \exists y \exists z \; (x \neq y \wedge y \neq z \wedge z \neq x)$

→ There is at most one cube.

$\forall x \forall y \; [(cube(x) \wedge cube(y)) \rightarrow (x=y)]$

OR.

No cube $\vee$ Exactly one cube

$\neg \exists x \; cube(x) \quad \vee \quad \exists x \forall y \; [cube(x) \wedge (cube(y) \rightarrow (x=y))]$

→ There is atmost one element

$\forall x \forall y \; (x=y)$

→ There is exactly one cube.

$\exists x \; (cube(x) \wedge \forall y \; (cube(y) \rightarrow (x=y)))$

$\equiv \exists x \forall y \; [cube(x) \wedge (cube(y) \rightarrow (x=y))]$

OR.

atleast one cube $\wedge$ atmost one cube.

$\exists x \; cube(x) \quad \wedge \quad \forall x \forall y \; [(cube(x) \wedge cube(y)) \rightarrow (x=y)]$

OR.

for ~~each~~ some cube x, all other cubes are only x.

i.e. $\exists x \forall y \; (cube(y) \leftrightarrow x=y)$

y is cube iff $x=y$.

→ There are at most two cubes

$\forall x \forall y \forall z \; [(cube(x) \wedge cube(y) \wedge cube(z)) \rightarrow ((x=y) \vee (y=z) \vee (z=x))]$

→ There are exactly two cubes

$\exists x \exists y \; [x \neq y \wedge \forall z \; (cube(z) \leftrightarrow [(z=x) \vee (z=y)])]$

$\exists x \exists y \; [(x \neq y) \wedge cube(x) \wedge cube(y) \wedge \forall z \; (cube(z) \rightarrow ((x=z) \vee (z=y)))]$

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→ For every no., there is a larger no.

$\forall x \exists y \ (x<y)$

→ There is a no. that is larger than every number

$\exists x \forall y \ (x>y)$

→ There is a no. that is larger than every other number

$\exists x \forall y \ [(x \neq y) \rightarrow (x>y)]$

→ Every tive no. is square

$\forall x \exists z \ [(x>0) \rightarrow (x*x=z)]$

→ If one no. is less than other no., then there is a no. properly b/w two.

• $\forall x \forall y \exists z \ [(x>y) \rightarrow ((x<z) \wedge (z<y))]$

→ There are atleast two prime no.

$\exists x \exists y \ [x \neq y \wedge prime(x) \wedge prime(y)]$

→ There are ∞ many prime no.

$\forall x \exists y \ [(y>x) \wedge prime(y)]$

i.e. for each prime no. there is a greater prime no. always.

→ a divides b.

$\exists x \ (ax=b)$

→ x is an even no.

$\exists y \ (2y=x)$

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→ n is prime.

$(n>1) \wedge \forall_y [y|n \rightarrow ((y=1) \vee (y=n))]$

i.e. n>1 & if y divides n then y is 1 or n.

→ relation "divides" is transitive. i.e a|b, b|c then a|c

$\forall_a \forall_b \forall_c [((a|b) \wedge (b|c)) \rightarrow (a|c)]$

→ There are an ∞ no. of pair of prime that differ by two (like (3,5), (5,7), (11,13), etc) (Twin prime conjecture)

$\forall n \exists y [(y>n) \wedge prime(y) \wedge prime(y+2)]$

→ All even no. greater than two are the sum of two primes. (Goldbach's conjecture)

$\forall n [(even(n) \wedge (n>2)) \longrightarrow \exists y \exists z (prime(y) \wedge prime(z) \wedge n=y+z)]$

★ **Negation of Quantifiers:**

→ S: Everyone has a car $\equiv \forall n [car(n)]$

¬S: Atleast one don't have car $\equiv \exists n [\neg car(n)] \equiv \neg \forall n [car(n)]$

→ M: Atleast one person has car $\equiv \exists n [car(n)]$

¬M: Nobody have car $\equiv \forall n [\neg car(n)] \equiv \neg \exists n [car(n)]$

★ So $\neg \forall n [p(n)] \equiv \exists n [\neg p(n)]$

& $\neg \exists n [p(n)] \equiv \forall n [\neg p(n)]$

So, ∃ changes to ∀ when neg. moves inside & vice versa.

$\neg \forall \equiv \exists \neg$ & $\neg \exists \equiv \forall \neg$

→ As, S: for every x, $p(x)$ is True
¬S: for some x, $p(x)$ is False

→ And S: for some x, $p(x)$ is True.
¬S: for all x, $p(x)$ is False.

→ Another way of understanding the same thing is:

a) $\forall x\ p(x) = p(a) \wedge p(b) \wedge \ldots \wedge p(n)$

$\neg \forall x\ p(x) = \neg(p(a) \wedge p(b) \wedge \ldots \wedge p(n))$

$= \neg p(a) \vee \neg p(b) \vee \neg p(c) \ldots \vee \neg p(n)$ (de morgan's law)

$= \exists x\ (\neg p(x))$

b). $\exists x\ p(x) = p(a) \vee p(b) \vee \ldots \vee p(n)$

$\neg \exists x\ p(x) = \neg(p(a) \vee p(b) \vee \ldots \vee p(n))$

$= \neg p(a) \wedge \neg p(b) \wedge \ldots \wedge \neg p(n)$ (de morgan law)

$= \forall x\ (\neg p(x))$

- Negation of $\forall x \exists y\ [p(x) \rightarrow q(y)]$ are:

a) $\neg \forall x \exists y\ [p(x) \rightarrow q(y)]$
b) $\exists x\ \neg \exists y\ [p(x) \rightarrow q(y)]$
c) $\exists x\ \forall y\ [\neg(p(x) \rightarrow q(y))]$
d) $\exists x\ \forall y\ [p(x) \wedge \neg q(y)]$

→ So, when negating nested quantifier, the negation can pass each quantifier left to right, turning it to the other kind.

★ Two useful equivalances

$\neg(p \wedge q) \equiv p \rightarrow \neg q$

$\neg(p \rightarrow q) \equiv p \wedge \neg q$

S: Every P is Q $\forall x\,[p(x) \rightarrow q(x)]$

¬S: Some P is not Q $\exists x\,[p \wedge \neg q]$

S: Some P is Q ~~All p are q~~ $\exists x\,(p(x) \wedge q(x))$

¬S: All P are not q $\forall x\,(p(x) \rightarrow \neg q(x))$

Ex: S: Everyone loves someone $\forall x\, \exists y$ loves(x,y)

¬S: There's someone who doesn't love anyone

$\neg\, \forall x\, \exists y$ loves(x,y)

$\exists x\, \neg \exists y$ loves(x,y)

$\exists x\, \forall y\, \neg$loves$(x,y)$

★ Validity and Satisfiability in FOL:

Valid = Always True (No false)

Satisfied = Atleast one True (Possible to be True).

So, let M be a First Order Logic Expression.

→ M is valid iff M is always True (whatever non empty domain we take and whatever predicate we take)

→ M is satisfiable iff M is possible to be True (for some domain & some predicate).

Ex: S: $\forall x\, p(x)$ is satisfiable as True for domain N and predicate $p(x): x > -1$. but not Valid because false for domain N & predicate $p(x)$: x is even.

Ex: S: $\exists x\, p(x)$ is also satisfiable & invalid bcz it can be sometime True & sometime false.

Ex: S: $\forall x \exists y\, p(x,y)$ is satisfiable as True for domain: N & predicate $p(x,y): x < y$.
It is also invalid as false for domain: N & predicate $p(x,y): x > y$ (No y for 1).

Que: Is it valid and satisfiable?
S: $\exists x\, Px \vee \exists y \sim Py$

Valid i.e. Always True.
Let $\alpha = \exists x\, px$ $\beta = \exists y(\sim py)$
S will be false if α = false & β = false.
α = false means $P(x)$ is not True for any
i.e. $P(x)$ is always false
So obviously β is True, bcz it says i.e.
at least one element is False.
So. $\alpha = F$ & $\beta = F$ is never possible
So S = Always True
= Valid

→ And if S is valid then it is satisfiable.
→ And if S is not satisfiable, then it is invalid

- valid → satisfiable (Reverse is not True)
- $\neg$Satisfiable → $\neg$Valid.

★ There are two ways to check the validity of FOL expression:

→ Method 1: Intution & Logical Thinking

→ Method 2: A systematic procedure:
- Take an abstract domain, like {a, b, c ...}
- Try to make expression, false some how:
  - If you can, it is invalid.
  - If you can never, it is valid.

★ To check satisfiablity think of a True example. It also has two ways:

→ Method 1: Intuition & Logical Thinking.

→ Method 2: A systematic procedure:
- Take an abstract domain, like {a, b, c, ... }
- Try to make exp. True some how:
  - If you can, it is satisfiable.
  - If you can't, it is not satisfiable.

NOTE: • In a FOL expression, by default, all expressions have same domain.

• Domain in FOL, is always Non-Empty, unless explicitly given as Empty.

Ans: Check validity & satisfiability:

a) S: $\forall x\, p(x) \wedge \exists y\, (\sim p(y))$

$\underbrace{\forall x\, p(x)}_{\alpha} \wedge \underbrace{\exists y\, (\sim p(y))}_{\beta}$

S can be True if $\alpha = T$ & $\beta = T$
but if $\alpha = T$, $\beta \neq T$ Obviously. So it is never True.

So unsatisfiable & so, invalid also.

→ Also $\forall x\, px \wedge \exists y \sim py \equiv \forall x\, px \wedge \exists x \sim px$
bcz, both variable are bounded (dummy).

b) S: $\exists x\, px \wedge \exists y \sim py$

S can be True if $\alpha = T$ & $\beta = T$.
It can be possible to be True, So satisfiable
It can be also false, so also Invalid
↳ if (a, b, c. all are True)
then $\alpha = T$ & $\beta = F \Rightarrow S = F$.

c) $\underbrace{(\forall x\, px)}_{\alpha} \rightarrow \underbrace{(\exists x\, px)}_{\beta}$

→ It can be F if $\alpha = T$ & $\beta = F$,
For $\alpha = T$ (all a, b, c are T) & Now $\beta = T$ always
so not possible To be False.
So it's valid & so obviously satisfiable.

This is bcz, we consider domain as non empty, o/w this would be Invalid, only satisfiable.

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d) $\forall x (px \lor \sim px)$

→ Always True, ⇒ Valid.

c) $(\forall x\ px) \lor (\forall x\ \sim px)$

→ Satisfiable only. (Invalid)

e) $\exists x (px \land \sim px)$

→ Not Satisfiable (Invalid) (Always false)

★ Validity of FOL expressions involving Implication $(\alpha \to \beta)$:

Try to make $\alpha \to \beta$ = false

For that $\alpha = T$ & $\beta = F$ simultaneously.

Use any approach (App 1, $\alpha = T \to \beta = F$ or App 2: $\beta = F \to \alpha = T$)

If we can, then it is invalid else Valid.

Ex:1 $\underbrace{\forall x (px \land qx)}_{\alpha} \to \underbrace{(\forall x\ px) \land (\forall x\ qx)}_{\beta}$

⇒ To check validity:

$\alpha = T$,

$x =$	a	b	c	— — —
px	T	T	T	—
qx	T	T	T	—

Now β can't be false as $\forall x\ px = T$ & $\forall x\ qx = T$

So, it is Valid.

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Ex:2 $\underbrace{(\forall x\, px) \wedge (\forall x\, qx)}_{\alpha} \longrightarrow \underbrace{\forall x\,(px \wedge qx)}_{\beta}$

→ Make $\alpha = T$

$x =$ a , b , c - - -
$px =$ T , T , T , - - -
$qx =$ T , T , T , - - -
$px \wedge qx =$ T , T , T , - - -

Now β cannot be made false, so it is Valid.

So as $\alpha \to \beta$ & $\beta \to \alpha$
$\alpha \equiv \beta$ or $(\alpha \leftrightarrow \beta)$

i.e. $\forall x\,(px \wedge qx) \equiv (\forall x\, px) \wedge (\forall x\, qx)$

Similarly we can prove.
$\forall x\,(px \wedge qx) \longrightarrow \forall y\, py$ (Valid)

but,
$\underbrace{\forall y\, py}_{\alpha} \not\rightarrow \underbrace{\forall x\,(px \wedge qx)}_{\beta}$ (Invalid)

$\alpha = T$

$x =$ a , b , c - -
$py =$ T , T , T .
$qy =$ F , F , F -

β is false & α is True so it is invalid.

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★ Equivalence in Predicate Logic (FOL): Two FOL exp. are logically equivalent iff, they have the same truth value for

- every predicate, &
- every domain

$S \equiv T$ iff $S \to T$ & $T \to S$.

Ex: $\forall x\, px \equiv \neg \exists x (\neg px)$

Que: $\underbrace{\forall x (px \vee qx)}_{\alpha} \overset{?}{\equiv} \underbrace{(\forall x (px)) \vee (\forall x (qx))}_{\beta}$

a) $\alpha \to \beta$.

→ Make $\alpha = T$

x	a	b	c	...
px	T	F	T	
qx	F	T	T	

$\beta = F$ is possible,

So $\alpha \to \beta$ is invalid

So $\alpha \not\equiv \beta$

Not Req. after a)

b) $\beta \to \alpha$

→ Make $\beta = T$

x	a	b	c	...
px	T	T	T	...
qx	T	F	T	

$\alpha = T$ Always

So valid, $\beta \to \alpha$ is valid.

Ex: Domain: N

px: x is Even

qx: x is odd

$\alpha = T$, $\beta = F$.

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Distributive Properties of Quantifiers:

Let $\Pi \rightarrow$ quantifier ($\exists$, $\forall$, $\exists!$)
& $\# \rightarrow$ connective ($\wedge, \vee, \oplus, \rightarrow, \leftrightarrow, \uparrow, \downarrow$)
then quantifier is distributive over connective iff

$$\underbrace{\Pi_x (p_x \# q_x)}_{\text{Compact form}} \equiv \underbrace{(\Pi_x p_x) \# (\Pi_x q_x)}_{\text{Expanded form}}$$

Now.

a). $\forall$ with $\wedge$: (Ex1 & Ex2 on prev page)

Compact $\rightleftarrows$ Expanded.

b) $\forall$ with $\vee$: (Prev. page)

Compact $\rightleftarrows$ Expanded

c) $\exists$ with $\wedge$: compact $\rightleftarrows$ Expanded

- $\exists x (\underbrace{p_x \wedge q_x}_{\alpha}) \rightarrow \underbrace{(\exists x p_x)}_{\beta} \wedge (\exists x q_x)$

if $\alpha = T$

$x =$ a , b , c ...

p_x : T , ...

q_x : T , ...

Now P can't be made false. So Valid.

- $\underbrace{\exists x\,(px \wedge qx)}_{\alpha} \leftarrow \underbrace{(\exists x\, px) \wedge (\exists x\, qx)}_{\beta}$

→ Make $\beta = T$

$x = a, b, c$ – –

$px = T$, – – – – –

$qx =$ – –, T – – –

α = Can be made false by taking all F.

So Invalid.

Similarly, we can do for any quantifier with any connective.
Here is a summarised table which we can by heart with a trick (Also we have understood):

$\rightarrow$ means Compact $\rightarrow$ Expanded

$\leftarrow$ means Compact $\leftarrow$ Expanded

$\leftrightarrow$ means Compact $\leftrightarrow$ Expanded

	$\forall$	$\exists$
$\wedge$	$\leftrightarrow$	$\rightarrow$
$\vee$	$\leftarrow$	$\leftrightarrow$
$\rightarrow$	$\rightarrow$	$\leftarrow$
$\leftrightarrow$	$\rightarrow$	×
$\oplus$	×	$\leftarrow$
$\uparrow$	$\rightarrow$	$\leftarrow$
$\downarrow$	$\rightarrow$	$\leftarrow$

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Way to byheart:

→ $\wedge$ & $\vee$, $\leftrightarrow$ & $\oplus$, $\uparrow$ & $\downarrow$ are all negation of each other so, is $\forall$ & $\exists$, so, their values will be opp. in case of $\wedge$ & $\vee$, $\leftrightarrow$ & $\oplus$, $\uparrow$ & $\downarrow$

→ $\forall$ with arrow types ($\uparrow, \downarrow, \leftrightarrow, \rightarrow$) is always $\rightarrow$.

→ $\forall$ with $\oplus$ is $\otimes$.

→ $\forall$ with $\wedge$ is $\leftrightarrow$ & with $\vee$ is $\leftarrow$.

Now with these tricks, can make the table. However we have also understood it. (we can prove whole table)

Que: which is valid (Tell from table)?

a) $[\forall n\, pn \rightarrow \forall n\, qn] \rightarrow \forall n (pn \rightarrow qn)$
→ Not valid ($\leftarrow$ was True)

b) $\forall n (pn \rightarrow qn) \rightarrow (\forall n\, pn \rightarrow \forall n\, qn)$
→ Valid.

c) $\exists n (pn \leftrightarrow qn) \rightarrow (\exists n\, pn \leftrightarrow \exists n\, qn)$
→ Invalid.

d) $\exists n (pn \uparrow qn) \rightarrow (\exists n\, pn \uparrow \exists n\, qn)$
→ Invalid.

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Que: Prove that

$$\underbrace{\forall x (px \rightarrow qx)}_{\alpha} \rightarrow \overbrace{(\forall x\, px \rightarrow \forall x\, qx)}^{\beta}$$

is valid.

→ Make $\alpha = T$

x	a	b	c	d	– – –
px	T	F	T	F	
qx	T	–	T	–	

$\beta = F$ is not possible bcz if $\forall x\, px$ is F, $\beta = T$
or if $\forall x\, px = T$, $\forall x\, qx$ will also be True, $\beta = T$.

Que: Check validity:

$$\underbrace{\forall x (px \leftrightarrow qx)}_{\alpha} \rightarrow \underbrace{\exists x\, px \oplus \forall x\, qx}_{\beta}$$

→ make $\alpha = T$

x	a	b	c	d	– –
px	T	T	T	T	
qx	T	T	T	T	

$\beta = F$ is possible if $\exists x\, px = T$ & $\forall x\, qx = T$
and for that all px & qx are True.

So Invalid

☆ Some standard book (Kenneth H. Rosen) Questions :

Q43 Determine if $\forall x(px \to qx)$ & $\forall x\, px \to \forall x\, qx$ are logically eq.

→ Let $\alpha = \forall x\,(px \to qx)$, & $\beta = (\forall x\, px \to \forall x qx)$

a) $\alpha \to \beta$

$\alpha =$ True.

x	a	b	c	— — —
px	T	F	T	
qx	T	F	T	

$\beta =$ False is not possible.

b) $\beta \to \alpha$

$\alpha =$ False

x	a	b	c	—
px	T	—	—	
qx	F	—	—	

$\beta =$ True is possible by making $\forall x\, px = F$.

So Not Equivalent.

Ques: $\forall x\,(px \oplus qx) \overset{?}{\equiv} (\exists x\, px) \to (\forall x qx)$

α (left), β (right)

→ $\alpha \to \beta$

$\alpha = T$

x	a	b	c	— —
px	T	F	T	— — —
qx	F	T	F	— — —

$\beta = F$ is not possible as $\exists x\, px$ is True & $\forall x\, qx$ is false.

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Tautology $\neq$ Validity (In FOL)

$\forall x\, A \equiv A$
$\exists x\, A \equiv A$ } if A has no free var x.

$\neg \forall x\,(px \wedge qx) \equiv \exists x[(\neg px) \vee (\neg qx)] \equiv \exists x(\neg px) \vee \exists x(\neg qx)$

★ **Null Quantification:** Some part which is unaffected by quantifier. (i.e. no free var. x)

Ex: $\forall x\,(px \wedge A)$
↳ A has no free var x

$\exists x\,(A \rightarrow px)$

Both are null quantification.

Null quantification rules are : (A little d/f from distributive)

WKT • $\forall x\,(px \wedge Ax) \not\equiv \forall x\, px \vee \forall x\, Ax$.

but,

$\underbrace{\forall x\,(px \vee A)}_{\alpha} \equiv \underbrace{\forall x\, px \vee \forall x\, A}_{\beta} \equiv \forall x\, px \vee A.$ — ①

A(x) has free var x & A have no free var x.

for proving eq ① we can make cases.

Case I: A = True
$\alpha = T$
$\beta = T$

A = false
$\alpha = \forall x\, px$
$\beta = \forall x\, px$

So equivalent.

Similarly

$(\exists x\, px) \vee A \equiv \exists x\,(px \vee A)$.

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- $\boxed{\begin{array}{l} \Pi_x (px \# A) \equiv (\Pi_x px) \# A. \\ \text{for } \Pi = \{\forall, \exists\} \\ \# = \{\vee, \wedge\} \end{array}}$ A don't have free variable x.

- $\forall x (px \to A) \not\equiv (\forall x\, px) \to A$
- $\exists x (px \to A) \not\equiv (\exists x px) \to A$
- $\forall x (A \to px) \equiv A \to (\forall x px)$
- $\exists x (A \to px) \equiv A \to (\exists x px)$

} Can be easily proved

(Learn by dialogue, if you want to go, go over my dead body, and consider A as dead)

- $\forall x (px \to A) \equiv (\exists x px) \to A.$
- $\exists x (px \to A) \equiv (\forall x\, px) \to A.$

} Can be proved (by taking cases)

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Ans: Which is not valid? (ω has no free var x)

A. $\forall x (px \vee \omega) \equiv (\forall x\, px) \vee \omega$

B. $\exists x (px \wedge \omega) \equiv (\exists x\, px) \wedge \omega$

C. $\forall x (px \to \omega) \equiv (\forall x px) \to \omega$ (Invalid)

D. $\forall x (px \to \omega) \equiv (\exists x px) \to \omega$

Made By - Karan Agrawal (GATE CS 2024 AIR 102)

★ Arguments in FOL: (Somewhat similar to that in prop. logic)

$$\left.\begin{array}{l} \left.\begin{array}{l} P_1 \\ P_2 \\ \vdots \\ P_n \end{array}\right\} \text{Premises} \\ \hline \therefore\ C \quad \} \text{ conclusion} \end{array}\right\} \text{Argument}$$

Arg. is valid iff whenever all premises are True, conclusion must be True.

i.e. $(P_1 \wedge P_2 \wedge \_\_ \wedge P_n) \rightarrow C$ is valid.

Ex: $\left.\begin{array}{l} \forall x\ p(x) \\ \hline \therefore\ p(b) \end{array}\right\}$ Valid Argument. $x \rightarrow \{a, b, c \_\_ \}$

Some Arguments:

a). Universal Instantiation:

$\left.\begin{array}{l} \forall x\ p(x) \\ \hline \therefore\ p(b) \end{array}\right\}$ valid, $x \in \{a, b, c \_\_ \}$

$\left.\begin{array}{l} p(b) \\ \hline \therefore\ \forall x\ p(x) \end{array}\right\}$ is invalid.

b). Universal Generalization:

$\left.\begin{array}{l} \text{for arbitrary } m,\ p(m) \\ \hline \therefore\ \forall x\ p(x) \end{array}\right\}$ Valid.

arbitrary means any $m \in$ domain.
Specific means particular element

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c) **Existential Instantiation**

$\exists x\ p(x)$
$\therefore$ for some particular $m \in$ domain, $p(m)$ } Valid.

$\exists x\ p(x)$
$\therefore\ p(a)$ } Invalid.

d) **Existential Generalisation**

$p(a)$
$\therefore\ \exists x\ p(x)$ } Valid.

$\exists x\ p(x)$
$\therefore$ for arb. m $p(m)$ } Invalid.

(These Names are not important).

e). $\forall x\ (p(x) \rightarrow q(x))$
$p(a)$, a is particular element
$\therefore\ q(a)$ } Valid
⇓
Similar to Modus Ponens of Prop. logic.

f). $\forall x\ (p(x) \rightarrow q(x))$
$\neg q(a)$, a is particular element
$\therefore\ \neg p(a)$ } Valid
⇓
Similar to Modus tollens of Prop. logic

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9). $\exists x\,(px \to qx)$
$p(a)$, a is particular element
$\therefore q(a)$
} Invalid.

Que: Show that the premises, "A student in this class has not read the book", & "Everyone in this class passed exam", implies the conclusion, "Some one who passed the exam has not read book".

→ If domain is All students in class

$\exists x\ \neg\,\text{Book}(x)$
$\forall x\ \text{pass}(x)$
$\therefore \exists x\,(\text{pass}(x) \wedge \neg\,\text{book}(x))$
} valid.

★ To check validity of Argument.

Method 1: (More used in FOL)	Method 2: (Used mostly in Prop. logic)
→ Make all premises True. ↓ If conclusion is Always True then valid, else Invalid	→ Make Conclusion: False ↓ If all premises are True, then Invalid, else Valid.

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Ex: 1: $\forall x\ (p(x) \lor q(x))$

2: $\forall x\ (s(x) \lor \neg q(x))$

3: $\forall x\ (R(x) \rightarrow \neg S(x))$

4: $\exists x\ (\neg p(x))$

$\therefore \exists x\ (\neg R(x))$

→ **Method 1:**

Make all premises True, start from Easy i.e. 4, 1, 2, 3

x	a	, b	, c	—
px	F			
qx	T			
Rx	F			
Sx	T			

Now conclusion is always True. (So valid)

→ **Method 2:**

Make conclusion false, and make all premises True.

x	a	, b	, c	,
px	T	, T	T	— —
qx	F	F	F	— —
Rx	T	T	T	— — —
Sx	F	F	F	— — —

premise 4 can't be made True, so Valid

⇒ **In FOL:**

To interpret $G: \forall x\ P(x)$ we need. a) $P(x)$

b) domain of x. (Non - Empty)

So, various Interpretation can be:

I1: Domain : N.

$P(x) = x$ is Even

Co model (G = False)

I2: Domain = N

$P(x) = x \geq 1$

model (G = True)

I3: Domain = {2, 4, 6}

$P(x) = x < 1$

Finite Co model (G = False)

I4: Domain = {2, 4, 6}

$P(x) = x$ is Even

finite model (G = True)

⇒ Interpretation with empty domain will not work.

☆ **Uniqueness Quantifier:** ($\exists!$) Used to represent exactly one.

Ex: Domain: Natural Number, $EP(x)$: Even prime x.

then $\exists!\ EP(x)$, means there is exactly one Even Prime.

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Exactly One = Unique = One & Only = ~~[illegible]~~ =
There is a Even prime x, and every even prime is same as x.

$\exists!_x (EP(x))$ is equivalent to

$$\exists x\, [EP(x) \land \forall y\, (EP(y) \to (x=y))]$$

$$\equiv \exists x\, [EP(x) \land \forall y\, ((x \neq y) \to \neg EP(y))]$$

$$\equiv \underbrace{\exists x\, (EP(x))}_{\text{(At least 1)}} \land \underbrace{\forall y \forall z\, (EP(y) \land EP(z) \to (y=z))}_{\text{(At most one)}}$$

$$\equiv \exists x \forall y\, (EP(y) \longleftrightarrow (y=x))$$

Ex: $p(x) = x+1=0$,
$\exists! x\ P(x)$ is True if domain is $\mathbb{Z}$, bcz -1 can only satisfy it.
$\exists! x\ p(x)$ is false if domain is $\mathbb{N}$, bcz no one can satisfy it.

$\Rightarrow$ **Negation of Uniqueness Identifier:**

$\exists! x\ p(x)$: Exactly one x satisfy $p(x)$

$\neg \exists! x\ p(x)$: No x satisfy OR
At least Two x satisfy $p(x)$

$$\neg \exists! x\ p(x) \equiv \neg \exists x\ p(x) \lor \exists y \exists z\ (p(y) \land p(z) \land y \neq z)$$

$$\equiv \forall x\ (\neg p(x) \lor \exists z\ (x \neq z \land p(z)))$$

i.e. for each x, either $\neg p(x)$ or there is $y \neq x$ and $p(y)$.

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$\exists x \rightarrow$ Atleast one (No. of Lovers)

$\exists! x \rightarrow$ Exactly one (No. of Mother)

$$\exists x\, p(x) \underset{\times}{\overset{\checkmark}{\rightleftarrows}} \exists! x\, p(x)$$

☆ Tautology in FOL:

→ In PL Tautology is same as Validity, but it is not the case in FOL.

→ Tautology is a concept of PL, not FOL. So we can't use our FOL knowledge to simplify expression, can only use PL knowledge. (It is expanded for FOL)

→ Tautology means Trivially True. So atomic prop is True but not Tautology.

For Ex: 2+2 = 4 is True Not Tautology, but
2+2 = 4 $\vee$ 2+2 ≠ 4 is Tautology.
$\underline{P \vee \neg P}$
↳ Trivially True

G: $\forall x\, p(x) \rightarrow \forall x\, p(x)$

Tautology Let $p = \forall x\, p(x)$
then $p \rightarrow p$ is Tautology.

but

G: $\forall x\, p(x) \rightarrow \neg \exists x\, \neg(p(x))$ (However Valid)

No Tautology bcz $p = \forall x\, p(x)$
$q = \neg \exists x\, \neg p(x)$
$p \rightarrow q$ is not Tautology (Can't use FOL knowledge.

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So, to know Tautology in FOL Expression, uniformly replace with prop. var.

→ Even if FOL statement is valid, we may not say it is tautology.

Ex: $G: \forall x\, p(x) \rightarrow \exists x\, p(x)$

G is valid (Always True),

but G is not Tautology. $(M \rightarrow N)$.

Que: Is following FOL exp. tautology?

$(\forall y \neg py \rightarrow \neg px) \rightarrow (px \rightarrow \neg \forall y \neg py)$

Let $M = \forall y \neg py$,

$N = px$

then

$(M \rightarrow \neg N) \rightarrow (N \rightarrow \neg M)$

It is tautology in PL, so Tautology

→ If FOL is Tautology, then surely it is Valid.

FOL Tautology $\underset{\times}{\overset{\checkmark}{\rightleftarrows}}$ Valid

PL Tautology $\underset{\checkmark}{\overset{\checkmark}{\rightleftarrows}}$ Valid

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Que: Which of the following WFF is a tautology.

A) $\forall x \exists y\, R(x,y) \leftrightarrow \exists y \forall x\, R(x,y)$

→ (Not Even valid) Not Tautology $(M \leftrightarrow N)$

B) $(\forall x (\exists y\, R(x,y) \neg S(x,y))) \rightarrow \forall x \exists y\, S(x,y)$

→ Not Tautology

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C). $\forall x \forall y\ P(x,y) \rightarrow \forall x \forall y\ P(y,x)$

→ Valid but not tautology $(M \rightarrow N)$

D). $[\forall x \exists y\ (P(x,y) \rightarrow R(x,y))] \rightarrow [\forall x \exists y\ (\neg P(x,y) \vee R(x,y))]$

→ Tautology (Use Prop. Logic Knowledge) $(P \rightarrow Q \equiv \neg P \vee Q)$

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For any questions or issues regarding my handwritten notes, feel free to reach out via email at karan757527@gmail.com or DM me on telegram at @karan757527 (or by clicking here).

I would love to know if you liked my notes. Your feedback and appreciation are invaluable, so don't hesitate to share your thoughts. Click here to connect with me on LinkedIn or you can find me as @agrawalkaran.

Wishing you success on your journey ahead!

- Karan Agrawal